

Attachment 3 — Current Status Snapshot

NLnet NGI Zero Commons Fund — Symphact application, 13th open call (deadline 2026-06-01).
Snapshot date: 2026-05-19. Repo: <https://github.com/FenySoft/Symphact>. Branch: `main`.

1. Milestone status at submission

Milestone	Status	Tests	Delivered
M0.1 Actor core primitives	✅ Shipped	30	<code>IMailbox</code> / <code>TMailbox</code> , <code>TActorRef</code> , <code>TActor<TState></code> , <code>TActorSystem</code>
M0.2 ActorContext + inter-actor messaging	✅ Shipped	+16	<code>IActorContext</code> / <code>TActorContext</code> , <code>DrainAsync</code> with <code>MaxRounds</code>
M0.3 Supervision (let-it-crash)	✅ Shipped	+30	<code>ISupervisorStrategy</code> , <code>TOneForOneStrategy</code> , lifecycle hooks, hierarchy
M0.4 Scheduler + per-actor parallelism	✅ Shipped	+86	<code>IScheduler</code> , <code>TInlineScheduler</code> , <code>TDedicatedThreadScheduler</code> , <code>IMailboxSignal</code> / <code>TDotNetMailboxSignal</code> , <code>QuiesceAsync</code>
M0.5 Persistence — BCL-only slices	✅ Shipped	+44	<code>IJournal</code> + <code>TInMemoryJournal</code> , <code>ISnapshotStore</code> + <code>TInMemorySnapshotStore</code>
M0.5 Persistence — content-addressed	🚧 In progress	—	<code>TCasJournal</code> + <code>TCasSnapshotStore</code> (SHA-256 content-addressed) — <i>grant deliverable M1</i>
M0.6 Remoting + capability registry	🕒 Planned	—	<code>ITransport</code> + <code>TTcpTransport</code> , <code>TCapabilityRegistry</code> — <i>grant deliverable M2</i>
M0.7 CFPU integration demo	🕒 Planned	—	End-to-end demo over <code>FenySoft.CilCpu.Sim</code> — <i>grant deliverable M4</i>

M0.3 supervision, M0.4 scheduler and the M0.5 BCL-only slices shipped between v0.1 (2026-04-16) and this submission, ~65 hours of self-funded TDD work. These are explicitly out of grant scope — the grant funds the content-addressed production-grade journal, supervision lifecycle integration, remoting, capability registry, kernel + device actors, CFPU integration, formal-verification groundwork, and outreach.

2. Test suite — 186 green xUnit tests

```
$ dotnet test --nologo
Passed! - Failed:    0, Passed:   120, Skipped:    0, Total:   120 - Symphact.Core.Tests.dll
Passed! - Failed:    0, Passed:    22, Skipped:    0, Total:    22 -
Symphact.Platform.DotNet.Tests.dll
Passed! - Failed:    0, Passed:    44, Skipped:    0, Total:    44 - Symphact.Persistence.Tests.dll
```

Test project	Count	Coverage
<code>Symphact.Core.Tests</code>	120	Actor core, scheduler, supervision, concurrency stress, mailbox signal
<code>Symphact.Platform.DotNet.Tests</code>	22	<code>TMailbox</code> (ConcurrentQueue MPMC), <code>TDotNetMailboxSignal</code> (AutoResetEvent WFI)
<code>Symphact.Persistence.Tests</code>	44	<code>TInMemoryJournal</code> , <code>TInMemorySnapshotStore</code> (append-only, replay, snapshot semantics)
Total	186	

CI runs the full suite on **Ubuntu / Windows / macOS** against .NET 10 SDK on every push (`.github/workflows/ci.yml`).

3. Code metrics

Layer	Files	Lines (.cs)
<code>src/Symphact.Core/</code>	18	runtime primitives, interfaces, supervision, scheduler
<code>src/Symphact.Persistence/</code>	6	journal + snapshot store interfaces + in-memory impls
<code>src/Symphact.Platform.DotNet/</code>	3	platform-specific .NET mailbox + signal impl
<code>src/Symphact.Platform.Cfpu/</code>	(stub)	awaiting CFPU F4 multi-core
src/ total	~27 files	~2 975 LOC
tests/ total	~20 files	~3 890 LOC
Test-to-runtime ratio		~1.31× (strict TDD discipline)

Build configuration (`Directory.Build.props`): `.NET 10`, `TreatWarningsAsErrors=true`, `GenerateDocumentationFile=true` — every warning is build-breaking, every public member requires XML docs.

4. Repository activity

- **Project start:** 2026-04-16 (initial repo scaffolding)
- **Snapshot date:** 2026-05-19
- **Calendar weeks:** ~5 weeks
- **Total commits:** 38
- **Focused TDD hours:** ~65 hours
- **Lines per hour:** ~106 (runtime + tests combined)

Recent commits (most recent first):

ac14a17 docs: NLnet pályázat v1.3 – post-Dennard paradigma-átírás + ördögügyvédi tényellenőrzés
 396cbd2 feat: M0.5 második szelet – ISnapshotStore + TInMemorySnapshotStore (BCL-only)
 1750e39 feat: M0.5 első szelet – IJournal + TInMemoryJournal (BCL-only)
 dc236c4 docs: „1 core = 1 actor” helytelen állítás javítása + roadmap szétválasztás
 db29cd1 docs: NLnet draft – PQC pontosítás LMS (NIST SP 800-208) hash-based signature-re
 fc44d9b docs: NLnet pályázati draft v1.1 – beadás-kész 13. nyílt körre
 e719fbe feat: M0.4 Scheduler + per-aktor parallelizmus
 8e61fc1 docs: Armstrong-Hewitt hivatkozások hitelességi javítása a vision dokumentumokban
 91eb5d0 docs: HMAC → CST modell átvezetés a Symphact dokumentációban
 5f5443c feat: M0.3 Supervision + HAL platform architektúra
 907bd46 feat: TActorRef(int SlotIndex) + CST capability modell – kaszkád átvezetés

5. Source-file inventory (runtime)

src/Symphact.Core/ (18 files):

ECoreStatus.cs	IPlatform.cs	TActorRef.cs
ESupervisorDirective.cs	IScheduler.cs	TActorSystem.cs
IActorContext.cs	ISchedulerHost.cs	TDedicatedThreadScheduler.cs
IMailbox.cs	ISupervisorStrategy.cs	TInlineScheduler.cs
IMailboxSignal.cs	TActor.cs	TOneForOneStrategy.cs
IMailboxSignalProvider.cs	TActorContext.cs	TTerminated.cs

src/Symphact.Persistence/ (6 files):

IJournal.cs	ISnapshotStore.cs	TInMemoryJournal.cs
TJournalEntry.cs	TSnapshotEntry.cs	TInMemorySnapshotStore.cs

src/Symphact.Platform.DotNet/ (3 files):

TDotNetMailboxSignal.cs	TDotNetPlatform.cs	TMailbox.cs
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6. Build verification

```
$ dotnet build Symphact.sln -c Debug
Symphact.Core          -> bin/Debug/net10.0/Symphact.Core.dll
Symphact.Persistence   -> bin/Debug/net10.0/Symphact.Persistence.dll
Symphact.Platform.DotNet -> bin/Debug/net10.0/Symphact.Platform.DotNet.dll
Symphact.Core.Tests    -> bin/Debug/net10.0/Symphact.Core.Tests.dll
Symphact.Persistence.Tests -> bin/Debug/net10.0/Symphact.Persistence.Tests.dll
Symphact.Platform.DotNet.Tests -> bin/Debug/net10.0/Symphact.Platform.DotNet.Tests.dll
Build succeeded.
    0 Warning(s)
    0 Error(s)
```

Zero warnings on a `TreatWarningsAsErrors=true` build with `GenerateDocumentationFile=true` — every public member is documented bilingually (`hu` / `en`) per repo convention.

7. Documentation state

Document	Version	Lines (en/hu)	Status
docs/vision-en.md / vision-hu.md	1.4	~750 / ~750	✓ Complete
docs/roadmap-en.md / roadmap-hu.md	1.0	~750 / ~750	✓ Complete
docs/trust-model-en.md / trust-model-hu.md	1.1	~250 / ~250	✓ Complete
docs/boot-sequence-en.md / boot-sequence-hu.md	—	—	✓ Complete
docs/ddr5-memory-model-en.md / ddr5-memory-model-hu.md	—	—	✓ Complete
docs/actor-ref-scaling-en.md / actor-ref-scaling-hu.md	—	—	✓ Complete
docs/osreq-to-cfpu/osreq-001..006	—	bilingual	✓ Active (007 superseded by CST model)
docs/nlnet-application-draft-en.md / -hu.md	1.3	302 / 350	✓ Submission-ready

Every document is bilingual (English + Hungarian) per repo convention. Cross-reference integrity is maintained across the two language sets.